

# EEG Sleep Classification and Reference-Based Sleep-Health Profiling

Biomarker-assisted N1/N2 refinement

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# Why detect sleep stages? Why this project?

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## Sleep supports memory, learning and physical health.

Duration alone does not show when sleep is interrupted or how stages are distributed.

### What stage detection gives us

Sleep onset and Wake time → continuity  
N1 / N2 / N3 / REM time → architecture  
A hypnogram → when patterns change

### Why we chose this project

EEG records brain activity used in staging.  
HMC provides labels to test our predictions.  
N1/N2 confusion is a measurable target.

## Potential benefit: assist sleep-study review and quantify patterns for research.

Our contribution: targeted refinement + descriptive profiles. Stage labels alone do not diagnose disease.

# Sleep stages and what a sleep cycle means

| Stage       | What it means                                         | Why we measure it                      |
|-------------|-------------------------------------------------------|----------------------------------------|
| <b>Wake</b> | Awake before or between sleep periods                 | Sleep onset and interruptions          |
| <b>N1</b>   | Transition from Wake to light sleep                   | Onset of sleep; distinguish N1 from N2 |
| <b>N2</b>   | Light non-REM sleep; brain activity slows             | Time in N2; our refinement target      |
| <b>N3</b>   | Deep, slow-wave sleep; supports restoration           | Time spent in deep sleep               |
| <b>REM</b>  | Active brain, rapid eye movements; dreaming is common | Timing and duration of REM sleep       |

**A cycle combines NREM and REM; these stages recur across the night.**

Typically, more N3 occurs early and REM periods lengthen later. Both phases support learning and memory.

# Literature review and project direction

| Study                                    | Main contribution                                        | Relevance to our project                                     |
|------------------------------------------|----------------------------------------------------------|--------------------------------------------------------------|
| DeepSleepNet<br>Supratak et al., 2017    | CNN features + BiLSTM<br>across EEG epochs               | Compare a learned<br>sequence-context baseline               |
| SeqSleepNet<br>Phan et al., 2019         | Hierarchical recurrent<br>sequence-to-sequence staging   | Add context between epochs;<br>current model is within-epoch |
| U-Sleep<br>Perslev et al., 2021          | Fully convolutional staging<br>across clinical cohorts   | Test transfer and robustness<br>on a separate dataset        |
| RobustSleepNet<br>Guillot & Thorey, 2021 | Montage flexibility and<br>leave-one-dataset-out testing | Check channel compatibility<br>and external generalization   |
| SLEEPYLAND<br>Dei Rossi et al., 2025*    | Standardized evaluation;<br>SOMNUS soft-voting ensemble  | Test model fusion and<br>calibrate the final output          |

**Design direction: targeted experts + sequence context + fair external evaluation**

\*2025 arXiv preprint. Published scores use different datasets and protocols; no direct ranking is claimed.

# Dataset and 30-second EEG epochs

HMC v1.1: 151 clinical PSG recordings; project subset: 24 recordings

| Partition  | Recordings | EEG epochs |
|------------|------------|------------|
| Training   | 16         | 15,216     |
| Validation | 4          | 3,619      |
| Test       | 4          | 3,109      |
| Total      | 24         | 21,944     |

One EEG epoch = one 30 s segment

0–30 s

30–60 s

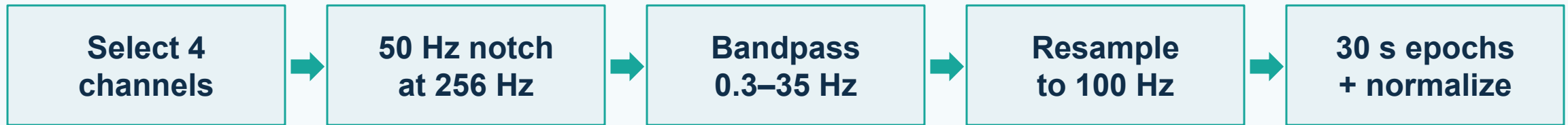
60–90 s

At 100 Hz:  $30 \times 100 = 3,000$  samples  
Four channels → input shape  $4 \times 3,000$

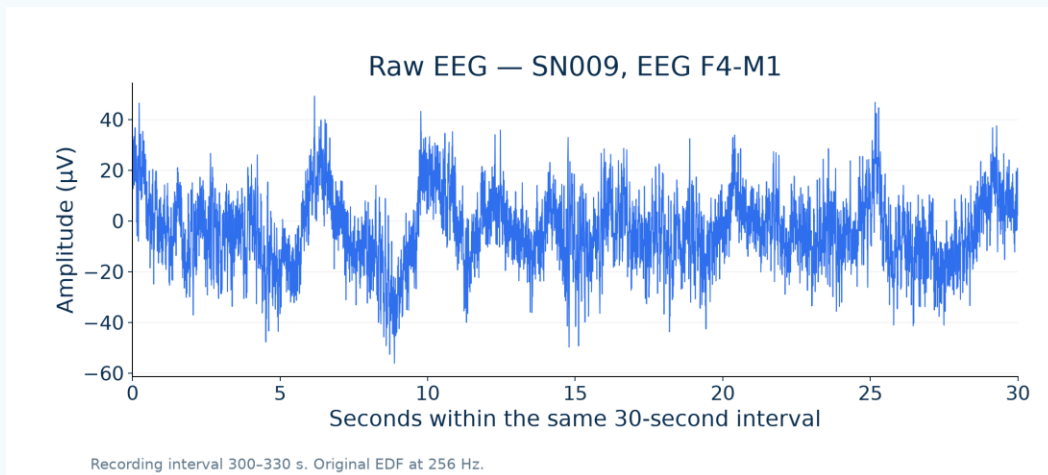
F4-M1 · C4-M1 · O2-M1 · C3-M2

Split by recording ID. SN001 covers only 190 epochs (95 min); whole-night claims are unsupported.

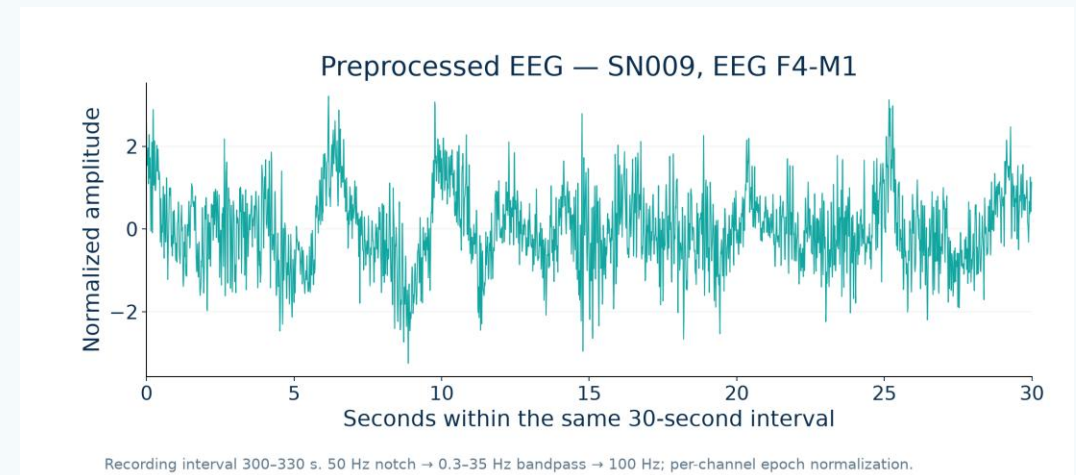
# EEG preprocessing



Raw EEG · SN009, F4-M1



Filtered and normalized EEG · same interval

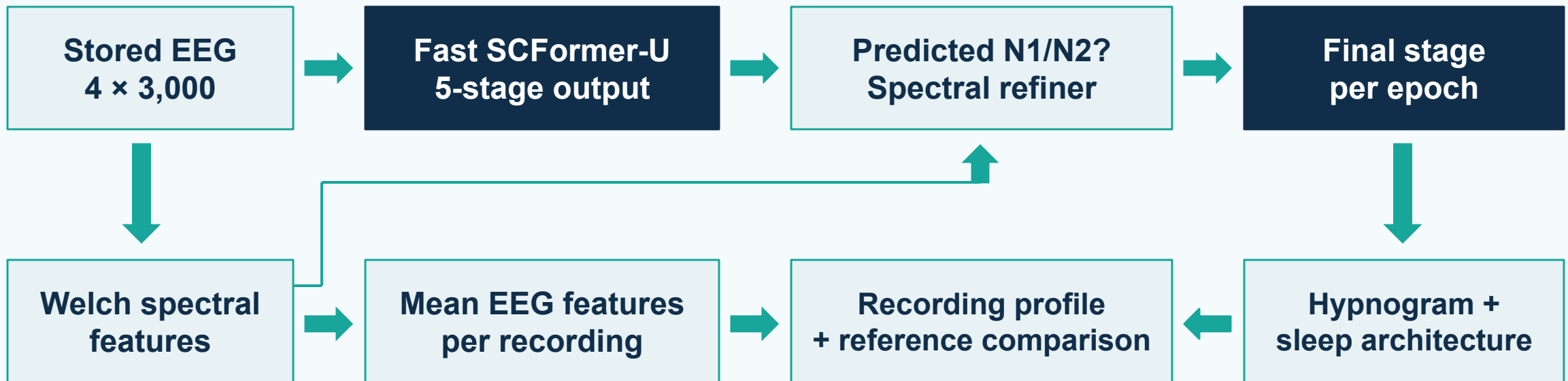


**Per-epoch, per-channel normalization. Apply the 50 Hz notch before resampling.**

Archived 300–330 s signal illustration only; all performance results use the current 3,109-epoch run.

# Implemented system block diagram

Main implementation: Biomarker\_N1\_N2\_Refinement.ipynb



Architecture + spectral features → z-scores / P10–P90 → four-domain profile → dashboard

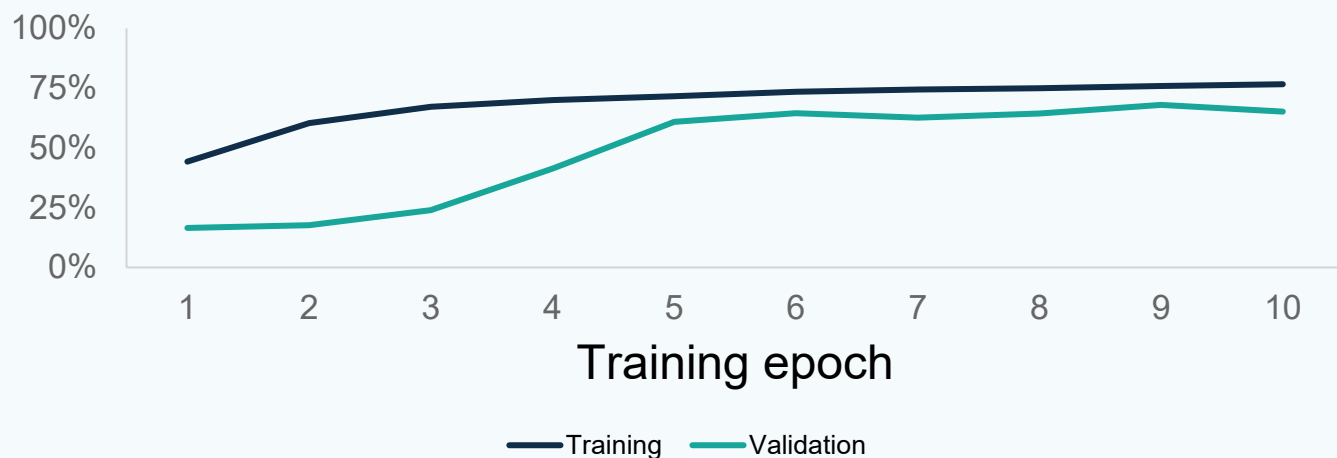
The 16-recording training reference provides dataset-relative descriptions.

# Existing model and training epochs



**226,309 parameters** · Attention and BiLSTM operate within each 30-second EEG epoch

## Training / validation accuracy



**A training epoch = one full pass through training data**

Adam · learning rate 0.0001

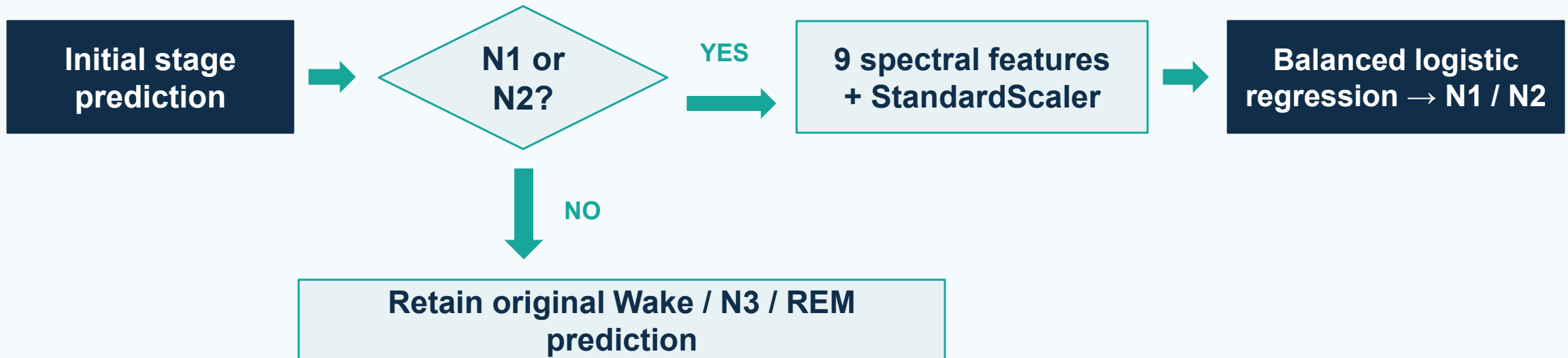
Batch size: 128

10 training epochs

Selected checkpoint: epoch 9

Current notebook loads this checkpoint; it trains the spectral refiner separately.

# N1/N2 refinement flowchart



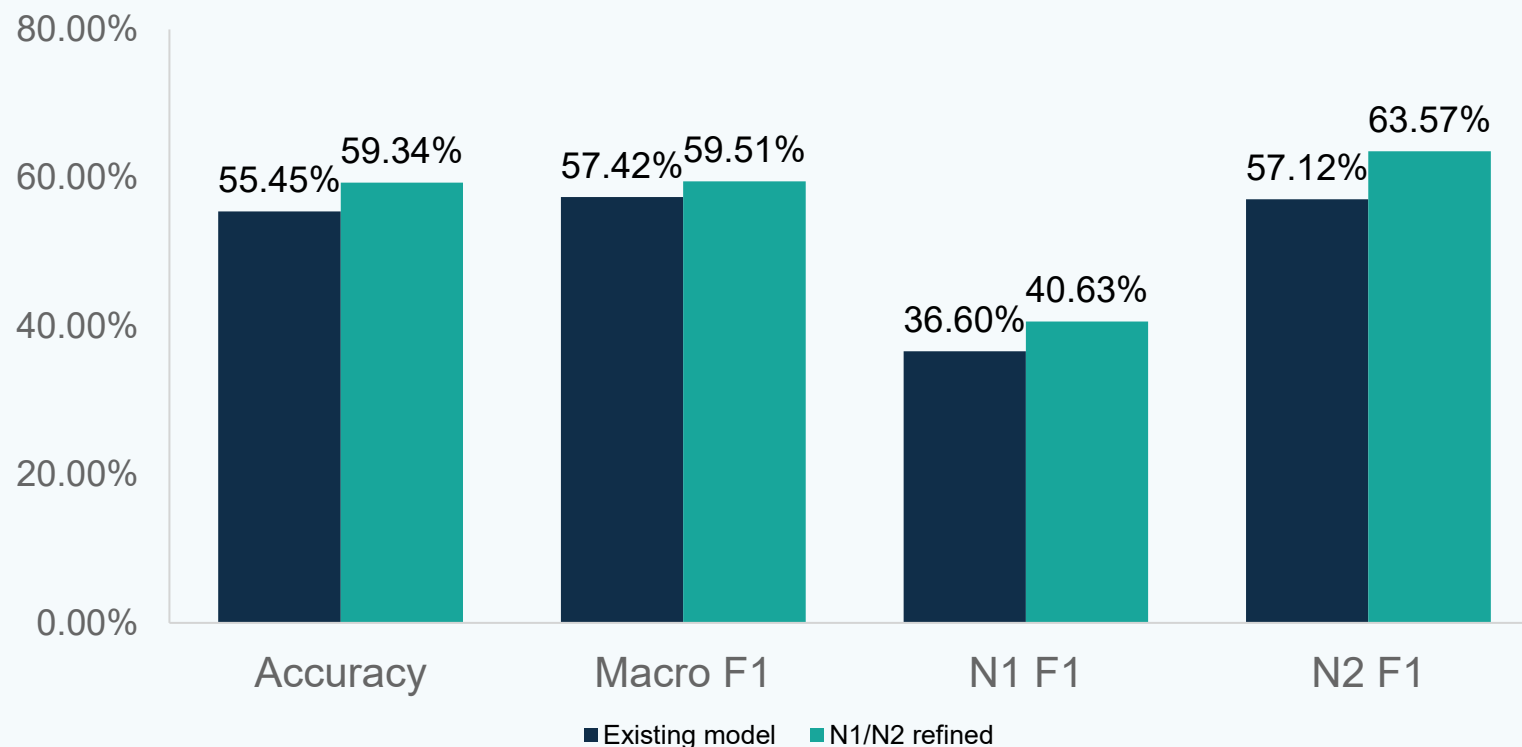
| Feature group       | Input to the refiner             |
|---------------------|----------------------------------|
| Relative powers     | Delta, Theta, Alpha, Sigma, Beta |
| Band / Delta ratios | Theta, Alpha, Sigma, Beta        |

## Fit on training data only

True AND predicted N1/N2:  
1,278 N1 + 4,588 N2 epochs  
Test gate uses predictions only.

Welch: 4 s windows, 2 s overlap; average channel powers before computing ratios.

# Results and measured gains



**+3.89 pp**

Accuracy: 55.45% → 59.34%

**+2.10 pp**

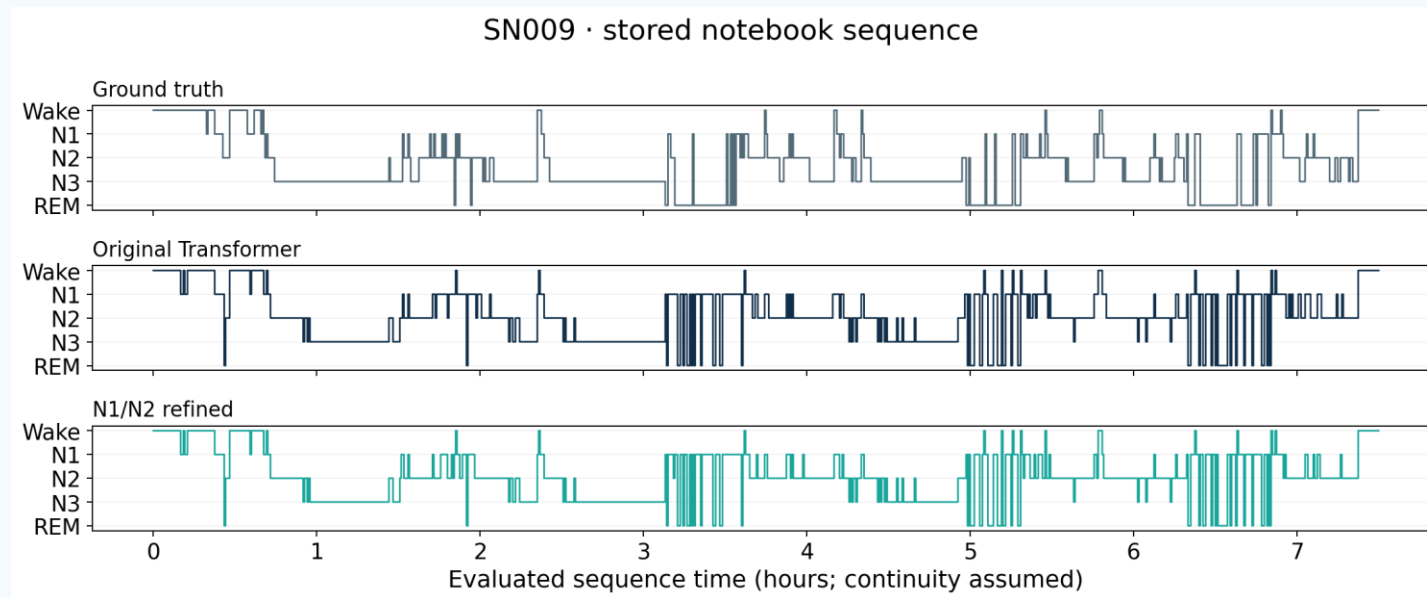
Macro F1:  
57.42% → 59.51%

**154 corrected – 33 degraded = 121 net additional correct epochs**

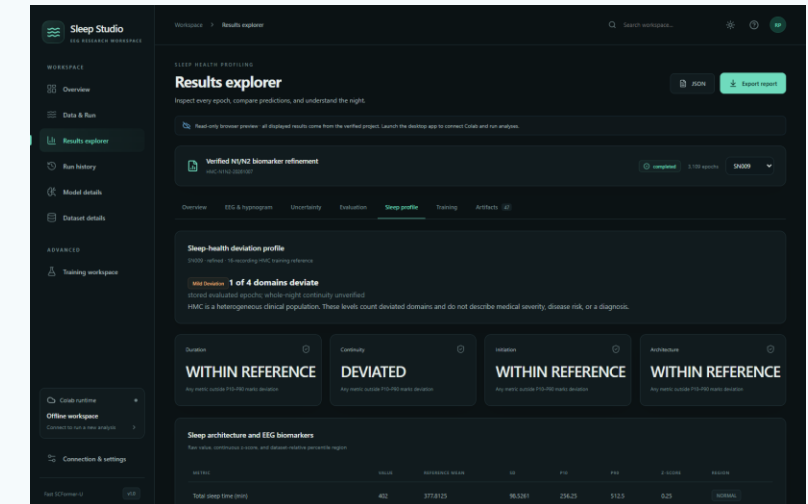
Same 3,109 test epochs across 4 recordings · 238 labels changed · pp = percentage points

# Project outputs: hypnogram and sleep profile

## SN009 · verified stage sequence



## Sleep Studio · current GUI



Architecture, spectral features,  
reference deviations and exports

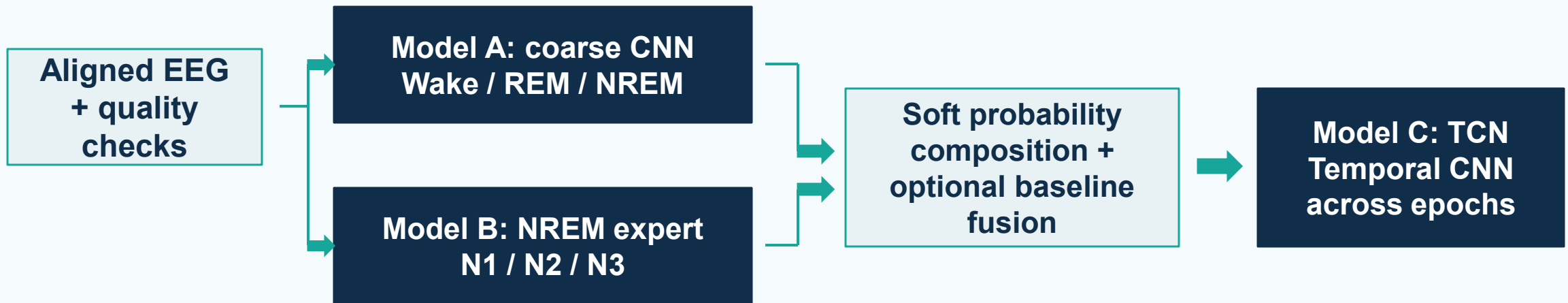
Stage labels → architecture metrics → training-reference comparison → four-domain profile

Duration · Continuity · Initiation · Architecture | Descriptive research profiles, not diagnoses

Notebook + verified run HMC-N1N2-20261007

# Future plan: multistage, multimodel architecture

Proposed • not yet implemented or evaluated



Conditional NREM probabilities preserve five-class normalization

## 1 Experts

Test coarse + NREM models against the current baseline.

## 2 Context + confidence

Train TCN on out-of-fold outputs. Calibrate final probabilities.

## 3 Broader validation

New locked cohort, then external data and optional EOG / EMG.

# Conclusion and next milestones

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## What is demonstrated

### **Spectral features improve N1/N2 discrimination in the current experiment.**

59.34% test accuracy; 59.51% macro F1.  
A complete notebook-to-GUI pipeline produces stage labels and profiles.

Limits: 4 test recordings, partial SN001,  
N1 F1 40.63%, no clinical validation.

## Next milestones

- 01 Audit full EDF alignment and quality; reserve a new locked test cohort.
- 02 Compare experts, soft fusion and TCN through controlled ablations.
- 03 Evaluate calibration, external data and biomarker errors; add modalities.

**Evaluate macro F1, N1/N2 F1, calibration, latency and errors in sleep biomarkers.**